CEFET/RJ

DAL & Comunidade

MODIFICAÇÃO DE VALORES

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Conteúdo

- Acesso a matrizes e tabela

Carregando baralho de uma URL

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
u <- "https://github.com/eogasawara/R/raw/main/baralho.rda"
load(url(u))
head(baralho)
```

Console

```
> u <- "https://github.com/eogasawara/R/raw/main/baralho.rda"
> load(url(u))
> head(baralho)
    face naipe valor
1     ás ouros     1
2   dois ouros     2
3   três ouros     3
4 quatro ouros     4
5  cinco ouros     5
6   seis ouros     6
>
```

Environment: baralho — 52 obs. of 3 variables; u — "https://github.com/eogasawar...

Files: C: > R > main — .RData 3.8 KB Oct 1, 2; .Rhistory 12 KB Oct 1, 2

Salvando baralho em arquivo

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```
1 save(baralho, file="baralho.rda")
2 rm(baralho)
3
```

3:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> save(baralho, file="baralho.rda")
> rm(baralho)
>
```

Environment History Connections Tutorial

Import | 143 MiB | List

R | Global Environment

values

u	"https://github.com/eogasawar...

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Salvando baralho em arquivo local

RStudio

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Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```
1 save(baralho, file="baralho.rda")
2 rm(baralho)
3
```

3:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> save(baralho, file="baralho.rda")
> rm(baralho)
>
```

Environment History Connections Tutorial

Import | 143 MiB | List

R | Global Environment

values

u	"https://github.com/eogasawar…

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

No Linux, deve-se executar save(baralho, file="baralho.rda", compress=TRUE)

Adicionando colunas a tabela

RStudio

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exemplo.R*

```
1  baralho$idx <- 1:52
2  head(baralho)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> baralho$idx <- 1:52
> head(baralho)
    face naipe valor idx
1     ás ouros     1   1
2   dois ouros     2   2
3   três ouros     3   3
4 quatro ouros     4   4
5  cinco ouros     5   5
6   seis ouros     6   6
>
```

Environment History Connections Tutorial

Data

baralho	52 obs. of 4 variables

Values

u	"https://github.com/eogasawar...

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Alterando valores vetores ou colunas

RStudio

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exemplo.R*

```r
baralho$idx[c(1,3,5)] <- 1
head(baralho)
baralho$idx[4:6] <- baralho$idx[4:6] + 1
head(baralho)
```

Console

```
> head(baralho)
    face naipe valor idx
1     ás ouros     1   1
2   dois ouros     2   2
3   três ouros     3   1
4 quatro ouros     4   4
5  cinco ouros     5   1
6   seis ouros     6   6
> baralho$idx[4:6] <- baralho$idx[4:6] + 1
> head(baralho)
    face naipe valor idx
1     ás ouros     1   1
2   dois ouros     2   2
3   três ouros     3   1
4 quatro ouros     4   5
```

Environment

Data: baralho — 52 obs. of 4 variables

Values: u — "https://github.com/eogasawar...

Filtrando dados

RStudio

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Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```r
baralho$idx <- 1:52
vec <- (baralho$idx %% 2 == 1)
idx <- baralho$idx[vec]
idx
cartas <- baralho[vec,]
cartas
```

7:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> baralho$idx <- 1:52
> vec <- (baralho$idx %% 2 == 1)
> idx <- baralho$idx[vec]
> idx
 [1]  1  3  5  7  9 11 13 15 17 19 21 23 25 27 29 31 33 35 37
[20] 39 41 43 45 47 49 51
> cartas <- baralho[vec,]
> cartas
     face  naipe valor idx
1      ás  ouros     1   1
3    três  ouros     3   3
5   cinco  ouros     5   5
7    sete  ouros     7   7
9    nove  ouros     9   9
11 valete  ouros    11  11
```

Environment History Connections Tutorial

Import | 118 MiB | List

R | Global Environment

baralho	52 obs. of 4 variables
cartas	26 obs. of 4 variables
Values	
idx	int [1:26] 1 3 5 7 9 11 13 15...
u	"https://github.com/eogasawar...
vec	logi [1:52] TRUE FALSE TRUE F...

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
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baralho.rda	451 B	Oct 1, 2

Operações de comparação

RStudio

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exemplo.R*

```r
1 > 2
1 > c(0, 1, 2)
c(1, 2, 3) == c(3, 2, 1)
1 %in% c(3, 4, 5)
c(1, 2, 3) %in% c(3, 4, 5)
```

Console

```
> 1 > 2
[1] FALSE
> 1 > c(0, 1, 2)
[1]  TRUE FALSE FALSE
> c(1, 2, 3) == c(3, 2, 1)
[1] FALSE  TRUE FALSE
> 1 %in% c(3, 4, 5)
[1] FALSE
> c(1, 2, 3) %in% c(3, 4, 5)
[1] FALSE FALSE  TRUE
>
```

Environment

baralho	52 obs. of 4 variables
cartas	26 obs. of 4 variables
Values	
idx	int [1:26] 1 3 5 7 9 11 13 15...
u	"https://github.com/eogasawar...
vec	logi [1:52] TRUE FALSE TRUE F...

Files

Name	Size	Modified
.RData	3.8 KB	Oct 1, 2
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baralho.rda	451 B	Oct 1, 2

Lembre-se dos demais operadores de comparação >, >=, <, <=, ==, !=

Operações de lógicos

RStudio

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exemplo.R*

```r
x <- baralho$face == "dama" & baralho$naipe == "espadas"
head(baralho[x,])

x <- baralho$face == "dama" | baralho$naipe == "espadas"
head(baralho[x,])
```

Console

```
> x <- baralho$face == "dama" & baralho$naipe == "espadas"
> head(baralho[x,])
   face   naipe
51 dama espadas
>
> x <- baralho$face == "dama" | baralho$naipe == "espadas"
> head(baralho[x,])
   face   naipe
12 dama   ouros
25 dama   copas
38 dama    paus
40   ás espadas
41 dois espadas
42 três espadas
>
```

Environment

```
    .. ..$ face : chr [1:13] "face=ás" "fa...
    .. ..$ naipe: chr [1:4] "naipe=ouros" ...
values
  idx     int [1:26] 1 3 5 7 9 11 13 15...
  u       "https://github.com/eogasawar...
  vec     logi [1:52] TRUE FALSE TRUE F...
  x       logi [1:52] FALSE FALSE FALSE...
```

Files

Name	Size	Modified
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

Lembre-se dos demais operadores de booleanos &, |, xor, !, any, all

Ausência de dados

RStudio

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exemplo.R*

```r
1 + NA
NA == 1
c(NA, 1:50)
mean(c(NA, 1:50))
mean(c(NA, 1:50), na.rm = TRUE)
```

Console — R 4.2.1 · C:/R/main/

```
> 1 + NA
[1] NA
> NA == 1
[1] NA
> c(NA, 1:50)
 [1] NA  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18
[20] 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37
[39] 38 39 40 41 42 43 44 45 46 47 48 49 50
> mean(c(NA, 1:50))
[1] NA
> mean(c(NA, 1:50), na.rm = TRUE)
[1] 25.5
>
```

Environment — Global Environment

```
.. ..$ face : chr [1:13] "face=ás" "fa...
.. ..$ naipe: chr [1:4] "naipe=ouros" ...
values
idx   int [1:26] 1 3 5 7 9 11 13 15...
u     "https://github.com/eogasawar...
vec   logi [1:52] TRUE FALSE TRUE F...
x     logi [1:52] FALSE FALSE FALSE...
```

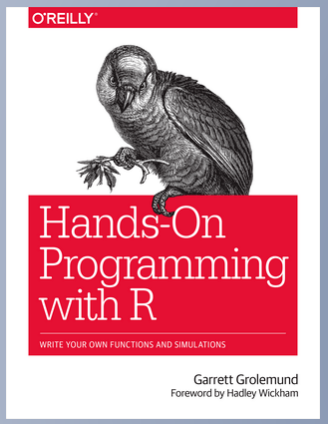
Files — C: > R > main

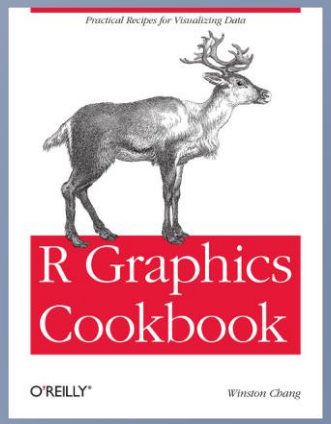
Name	Size	Modified
..		
.RData	3.8 KB	Oct 1, 2
.Rhistory	12 KB	Oct 1, 2
baralho.rda	451 B	Oct 1, 2

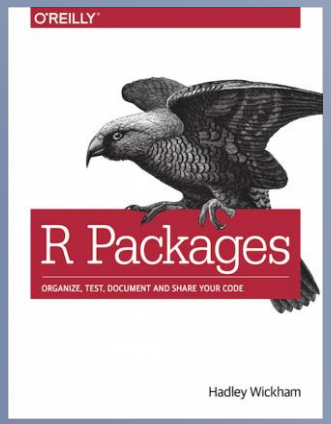
Lembre-se dos demais operadores de booleanos &, |, xor, !, any, all

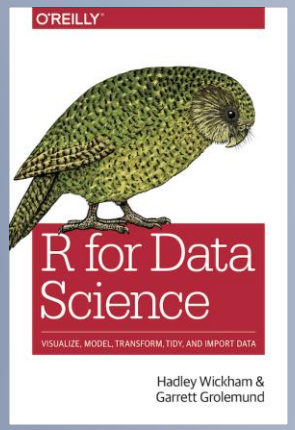
Referências

Código: https://github.com/eogasawara/R/blob/main/examples/04-Notacao.md









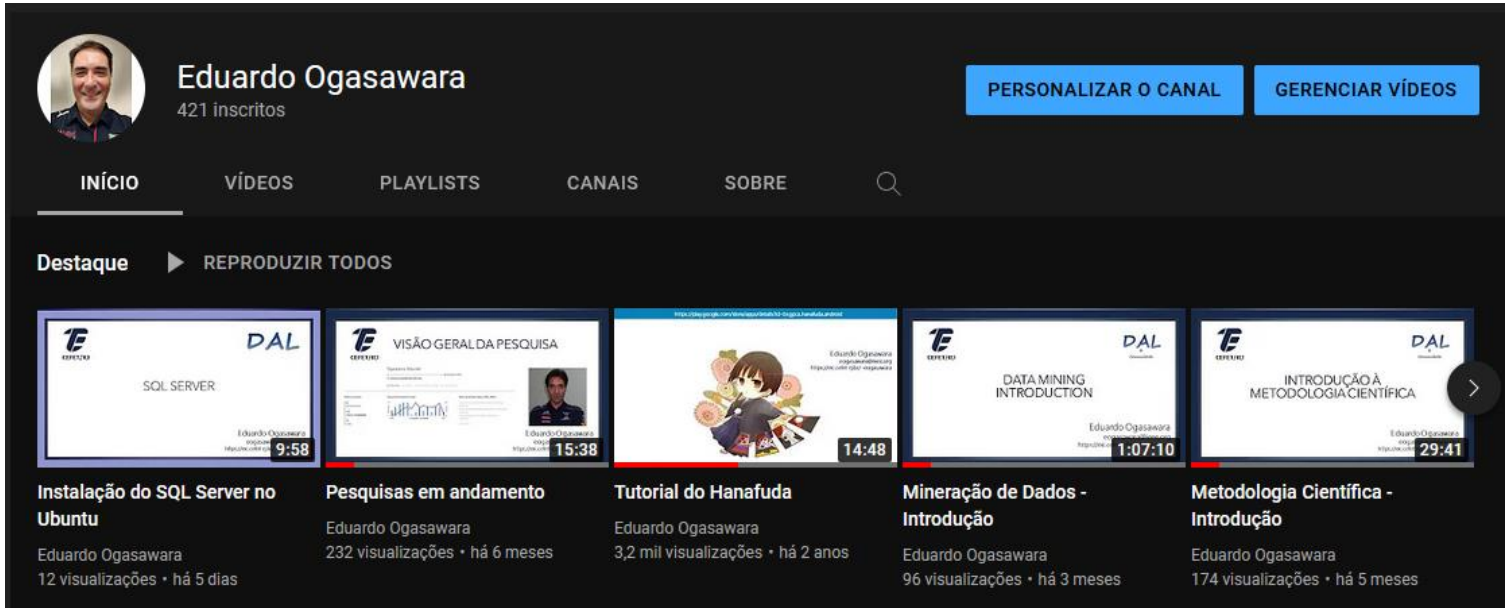
Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

Leitura adicional: https://rstudio-education.github.io/hopr/r-notation.html

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

Mantenha-se atualizado

Inscreva-se: https://eic.cefet-rj.br/~eogasawara/youtube



Slides and Jupyter notebooks: https://github.com/eogasawara/R